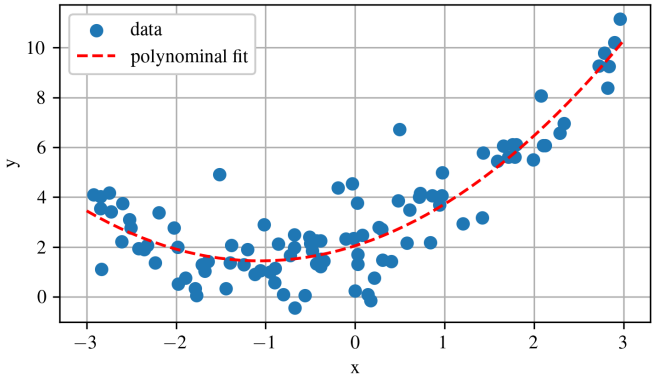
Chapter 2 Regression

Austin R.J. Downey

Regression



Regression models the relationship between an input variable x (data) and an output variable y (target). The goal is to learn a function that predicts y from x.

Regression Approaches

In this chapter, we examine Linear Regression and two ways to estimate parameters:

- **Closed-form solution:** Solve directly for parameters by minimizing the cost function.
- **Gradient Descent:** Iteratively update parameters to reduce error.

Gradient Descent variants:

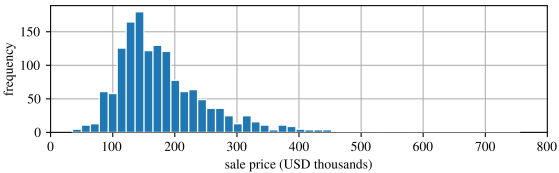
- Batch Gradient Descent
- Stochastic Gradient Descent
- Mini-batch Gradient Descent

Closed-form gives an exact solution immediately, while Gradient Descent converges through successive updates.

Data: Ames Housing Dataset

The Ames housing dataset contains:

- 2,930 home sales
- Data from 2006–2010
- Multiple feature types

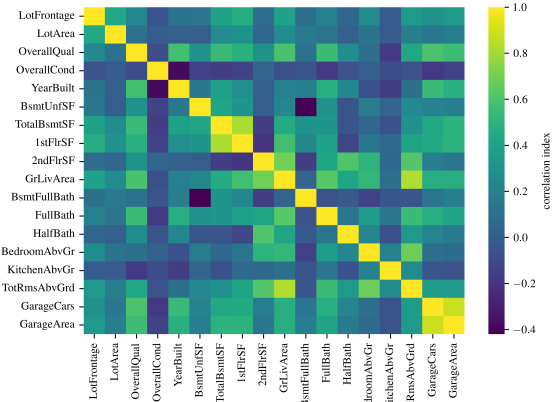


Histogram of sale prices.

Data: Ames Housing Dataset Feature Correlations

Feature relationships:

- Shows correlations between inputs
- Helps identify important variables
- Useful for feature selection

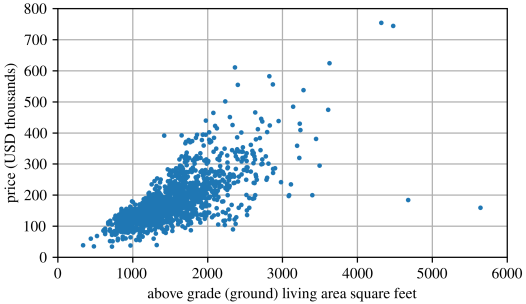


Correlation matrix of features.

Ames Housing Example

We want to predict house price using the Ames dataset.

- Input: above-ground living area
- Target: housing price
- Goal: find a linear relationship between them



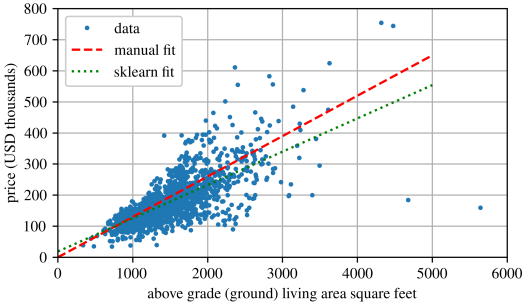
Ames housing data.

A Simple Model

A clear trend is visible: larger homes tend to cost more.

$$\text{price\_model} = \theta_1 + \theta_2 \times \text{above\_ground\_living\_area}$$

- θ_1 is the intercept
- θ_2 is the slope
- The model can represent any line



Ames housing data with trend lines.

Notation

Notations

In terms of linear algebra notation; we use:

- $\mathbf{x}$: lowercase bold symbols denote vectors.
- $\mathbf{X}$: uppercase bold symbols denote matrices.

In machine learning, we commonly use:

- $\mathbf{X}$: the feature matrix (input data).
- $\mathbf{y}$: the target vector (labels or outputs).
- $\mathbf{x}^{(i)}$: the i^{th} instance in the dataset.
- h: the hypothesis or prediction function, $\hat{\mathbf{y}} = h(\mathbf{X})$.
- n: the number of features.
- m: the number of training instances.
- $\hat{x}$: an estimated value ("x-hat").

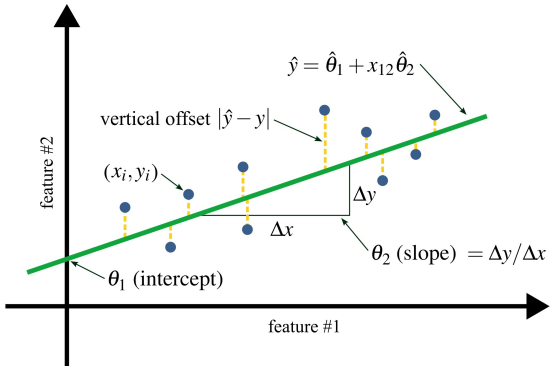
Least Squares Intuition

Goal:

- Find parameters that minimize error
- Fit a line closest to all data points

Idea:

- Measure vertical distance to line
- Square the errors
- Sum over all points



Least squares fit.

How Do We Measure Fit?

Before tuning a model, we need a way to measure how well it fits the data.

- Define a **cost function**
- Penalizes differences between predictions and true values
- Lower cost = better fit

For linear regression:

- Use **Ordinary Least Squares (OLS)**
- Minimizes prediction error over all training data

Mean Squared Error (MSE)

A common cost function is Mean Squared Error:

$$J = \frac{1}{m} \sum_{i=1}^{m} (\hat{y}_i - y_i)^2$$

- m: number of data points
- Measures average squared prediction error
- Larger errors are penalized more heavily

For linear regression:

$$J = \frac{1}{m} \sum_{i=1}^{m} (\boldsymbol{\theta}^\top \mathbf{x}^{(i)} - y^{(i)})^2$$

Linear Regression Model

Consider a dataset with n observations and p features.

For each observation:

$$y_i = \theta_1 x_{i1} + \theta_2 x_{i2} + \cdots + \theta_p x_{ip} + \epsilon_i$$

- y_i: target value
- x_{ij}: feature values
- θ_j: model parameters
- ϵ_i: error term

Matrix Formulation

We can write the linear model compactly:

$$\mathbf{y} = \mathbf{X}\boldsymbol{\theta} + \epsilon$$

- $\mathbf{X}$: feature matrix
- $\boldsymbol{\theta}$: parameters
- $\mathbf{y}$: target values

Predictions:

$$\hat{\mathbf{y}} = \mathbf{X}\hat{\boldsymbol{\theta}}$$

Goal:

- Find $\hat{\boldsymbol{\theta}}$ that minimizes error

$$\mathbf{X} = \begin{bmatrix} x_{11} & x_{12} & \cdots & x_{1p} \\ x_{21} & x_{22} & \cdots & x_{2p} \\ \vdots & \vdots & \ddots & \vdots \\ x_{n1} & x_{n2} & \cdots & x_{np} \end{bmatrix}$$

$$\boldsymbol{\theta} = \begin{bmatrix} \theta_1 \\ \theta_2 \\ \vdots \\ \theta_p \end{bmatrix} \qquad \mathbf{y} = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix}$$

The Normal Equation

Closed-form solution:

$$\hat{\boldsymbol{\theta}} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}$$

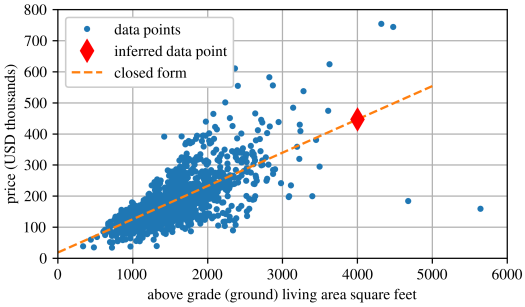
- Gives the exact best-fit parameters
- No iteration required
- Works well for small feature sets

For a simple model:

$$\hat{y} = \hat{\theta}_1 + x\hat{\theta}_2$$

This is just:

$$y = mx + b$$



Ames housing data inferred.

Computational Complexity

- Normal Equation requires inverting $\mathbf{X}^\top \mathbf{X}$ (an $n \times n$ matrix)
- Matrix inversion cost: $O(n^{2.4})$ to $O(n^3)$
- Doubling features $\Rightarrow \sim 5$–$8\times$ more computation

Warning

For very large feature sets (e.g., $n \sim 100{,}000$), the Normal Equation becomes extremely slow.

- Scales linearly with number of instances: $O(m)$
- Prediction is fast: linear in both m and n

Example 2.1: Ames Housing (Linear Regression)

- Fit a linear model to predict house price from living area
- Compare approaches:
  - Manual fit (visual intuition)
  - Closed-form (Normal Equation)
  - Gradient Descent
- In practice (scikit-learn):
  - `LinearRegression` (closed-form)
  - `SGDRegressor` (iterative)

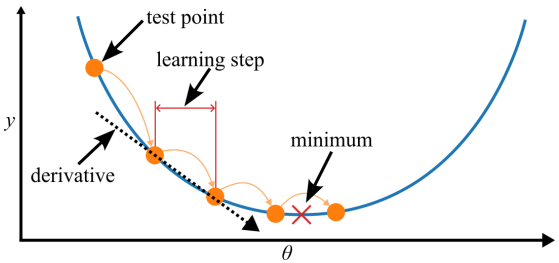
Next: methods that scale better for large feature sets or large datasets.

Gradient Descent

- Optimization algorithm to minimize a cost function
- Iteratively updates parameters

Idea:

1. Compute the gradient (slope)
2. Move in direction of steepest descent
3. Repeat until minimum is reached

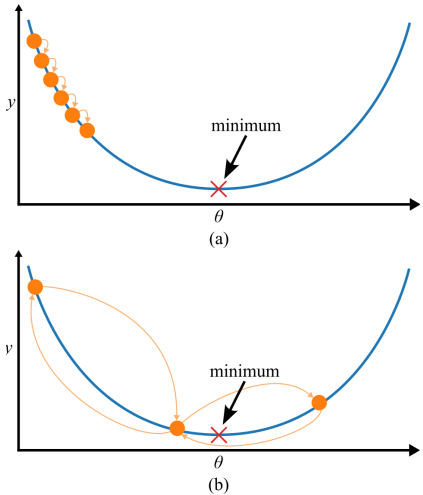


Gradient descent moving toward a minimum.

Learning Rate

- Step size controlled by learning rate η
- Small $\eta \rightarrow$ slow convergence
- Large $\eta \rightarrow$ overshooting / divergence

Choosing η is critical



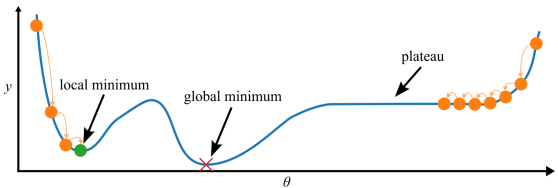
Effect of different learning rates.

Challenges

- Cost functions may be complex:
  - Local minima
  - Plateaus
  - Ridges
- Starting point matters
- Convergence may be slow

For linear regression:

- MSE is convex
- One global minimum

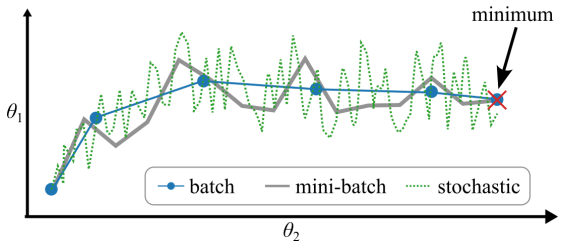


Optimization challenges in complex landscapes.

Comparing Gradient Descent Methods

- Different methods reach the minimum differently
- Trade-off between speed and stability
- Some methods are noisy but fast
- Others are stable but computationally expensive

All aim to minimize the same cost function



Paths taken by different methods.

Method Comparison

Different approaches to linear regression vary in speed, scalability, and flexibility.

algorithm	large m	in memory	large n	hyperparams	scaling	sklearn
normal equation	fast	yes	slow	0	no	`LinearRegression`
batch GD	slow	yes	fast	2	yes	n/a
stochastic GD	fast	no	fast	≥ 2	yes	`SGDRegressor`
mini-batch GD	fast	no	fast	≥ 2	yes	n/a

NOTE

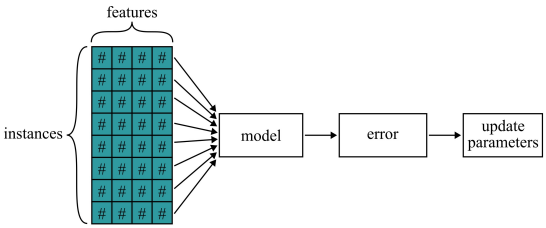
After training, there is minimal difference between the models: these algorithms converge on similar solutions and make predictions in a nearly identical manner.

Batch Gradient Descent

- Uses the entire dataset at each step
- Minimizes the cost function over all data
- Based on Mean Squared Error (MSE)

$$J = \frac{1}{m} \sum_{i=1}^{m} (\boldsymbol{\theta}^\top \mathbf{x}^{(i)} - y^{(i)})^2$$

- Computes exact gradient each iteration
- Can be slow for large datasets



Batch Gradient Descent flow.

Gradient Computation

Compute partial derivatives:

$$\frac{\partial}{\partial \theta_j} J(\theta) = \frac{2}{m} \sum_{i=1}^{m} (\boldsymbol{\theta}^\top \mathbf{x}^{(i)} - y^{(i)}) x_j^{(i)}$$

Vector form:

$$\nabla_\theta J(\boldsymbol{\theta}) = \frac{2}{m} \mathbf{X}^\top (\mathbf{X}\boldsymbol{\theta} - \mathbf{y})$$

- Computes all slopes at once
- Direction of steepest ascent

Key Idea

Move in the opposite direction of the gradient to reduce error.

Warning

Uses the full dataset at every step → can be slow for large m.

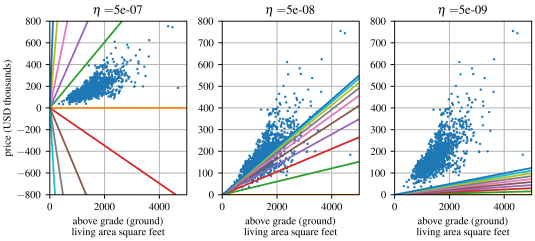
Parameter Update and Learning Rate

Update rule:

$$\boldsymbol{\theta}^{(\text{next})} = \boldsymbol{\theta} - \eta \nabla_{\theta} J(\boldsymbol{\theta})$$

- η: learning rate (step size)
- Controls speed of convergence

- Too large → divergence
- Too small → slow convergence
- Just right → fast convergence

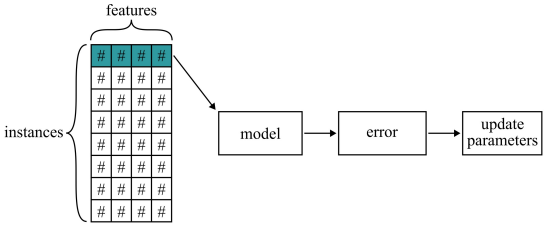


Effect of different learning rates.

Stochastic Gradient Descent

- Updates parameters using one sample at a time
- Uses $(x^{(i)}, y^{(i)})$ instead of full dataset

- Faster for large datasets
- Enables online learning



Flow of Stochastic Gradient Descent.

SGD Update Rule

Update rule:

$$\boldsymbol{\theta}^{(\text{next})} = \boldsymbol{\theta} - \eta \nabla_{\theta} J(\boldsymbol{\theta}; x^{(i)}, y^{(i)})$$

- Updates after every single example
- Avoids redundant computations
- More efficient than Batch GD for large m

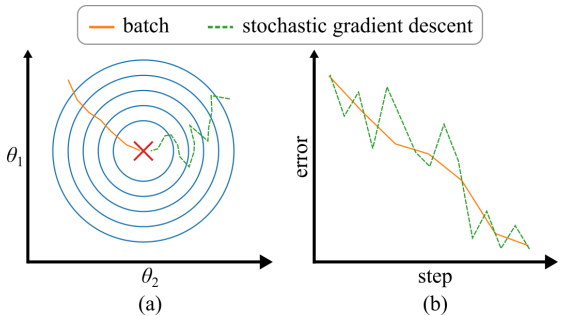
Key Idea

Instead of waiting to see all data, update immediately using each new sample.

SGD Behavior

- Fast updates, but noisy path
- High variance in cost function
- Does not decrease smoothly

- Often oscillates around minimum
- Can still reach a good solution



Batch vs. stochastic gradient descent behavior.

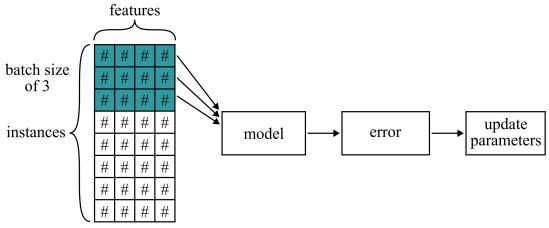
Mini-batch Gradient Descent

- Uses small random subsets (mini-batches)
- Between Batch GD and Stochastic GD

Update rule:

$$\boldsymbol{\theta}^{(\text{next})} = \boldsymbol{\theta} - \eta \nabla_\theta J(\boldsymbol{\theta}; x^{(i:i+n)}, y^{(i:i+n)})$$

- Faster than Batch GD
- More stable than SGD
- Efficient on GPUs (matrix operations)

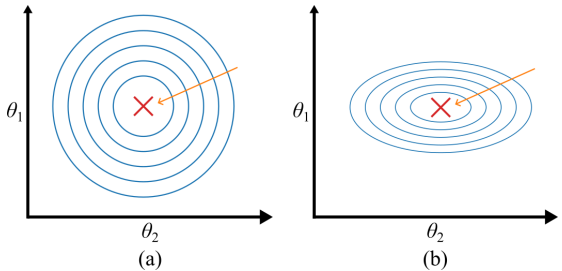


Flow of Mini-batch Gradient Descent.

Feature Scaling

- Different feature scales distort the cost surface
- Creates an elongated “valley”
- Equal scaling → fast convergence
- Unequal scaling → slow convergence

Gradient Descent becomes inefficient without scaling



Effect of feature scaling on convergence.

Scaling Methods

Min-max scaling

$$\mathbf{X}' = \frac{\mathbf{X} - \mathbf{X}_{\min}}{\mathbf{X}_{\max} - \mathbf{X}_{\min}}$$

- Rescales to $[0, 1]$
- Sensitive to outliers

Standardization

$$\mathbf{X}' = \frac{\mathbf{X} - \overline{\mathbf{X}}}{\sigma}$$

- Zero mean, unit variance
- Less sensitive to outliers

- Both improve Gradient Descent performance
- Standardization is most commonly used

Tools

`MinMaxScaler`
`StandardScaler`

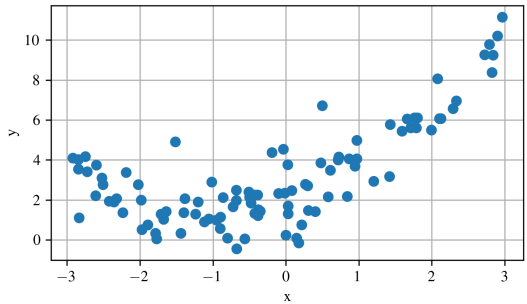
Polynomial Regression

What if the data is not linear?

A quadratic model can represent curved behavior:

$$\hat{y} = ax^2 + bx + c$$

- Linear models underfit nonlinear data
- Polynomial terms allow curved fits
- Still linear in the parameters



Dataset generated from a quadratic relationship.

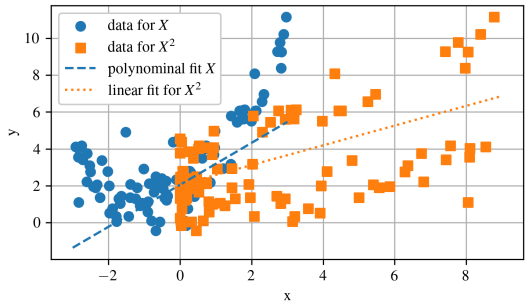
Feature Expansion

Polynomial regression works by expanding the feature space.

For example:

$$x \to [x,\ x^2]$$

- Add higher-order terms as new features
- Train a linear model on the expanded data
- Allows nonlinear behavior



Linear fits to the expanded feature set.

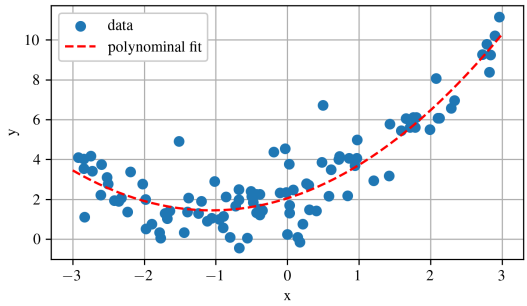
Polynomial Model Fit

The learned coefficients combine into:

$$\hat{y} = ax^2 + bx + c$$

- a: quadratic term
- b: linear term
- c: bias/intercept

The resulting model can approximate curved data.



Polynomial regression fit.

Polynomial Features

Polynomial regression can also model feature interactions.

Example with two features a and b:

$$[a,\ b]$$

Degree-3 expansion:

$$[a,\ b,\ a^2,\ b^2,\ a^3,\ b^3,\ ab,\ a^2b,\ ab^2]$$

- Captures nonlinear relationships
- Captures interactions between variables
- Implemented with:

```
PolynomialFeatures(degree=d)
```

Combinatorial Explosion

Warning

Polynomial feature expansion can rapidly increase the number of features.

Number of expanded features:

$$\frac{(n+d)!}{d!n!}$$

- n: original number of features
- d: polynomial degree

Higher polynomial degrees can:

- Increase memory usage
- Slow training
- Cause overfitting

Example 2.2: Polynomial Regression

- Fit a nonlinear dataset using polynomial regression
- Add x^2 as an additional feature
- Train a linear model on the expanded feature set

Demonstrates how:

- Linear models can approximate curved relationships
- Polynomial terms improve model flexibility
- The resulting fit compares to the original data

Tools used:

`PolynomialFeatures` `LinearRegression`